Multithreaded Architectures (cont.) and Programming on Multithreaded Architectures: Never say never

Hung-Wei Tseng

Recap: Tips of programming on modern processors

- Minimize the critical path operations
  - Don’t forget about optimizing cache/memory locality first!
    - Memory latencies are still way longer than any arithmetic instruction
    - Can we use arrays/hash tables instead of lists?
  - Branch can be expensive as pipeline get deeper
    - Sorting
    - Loop unrolling
  - Still need to carefully avoid long latency operations (e.g., mod)
- Since processors have multiple functional units — code must be able to exploit instruction-level parallelism
  - Hide as many instructions as possible under the “critical path”
  - Try to use as many different functional units simultaneously as possible
- Modern processors also have accelerated instructions
- Compiler can do fairly go optimizations, but with limitations

Recap: limited ILP in programs

	Cycles	IC	IPC/ILP	ET	# of branches	Branch mis-prediction rate
A	334270949858	118508036769	2.821	23.237	65366730559	0.012
B	290179950840	68341242029	4.246	13.419	18319495534	0.004
C	102882218758	27580097715	3.730	5.380	18129328282	0.004
D	80043714833	19072124536	4.197	3.754	1037120144	0.000
E	253238690876	376641071475	0.672	73.977	73967642413	0.184
SSE4.2	22089754243	8444815558	2.616	1.654	1014543318	0.000

Most of time, we can't fully utilize the function units

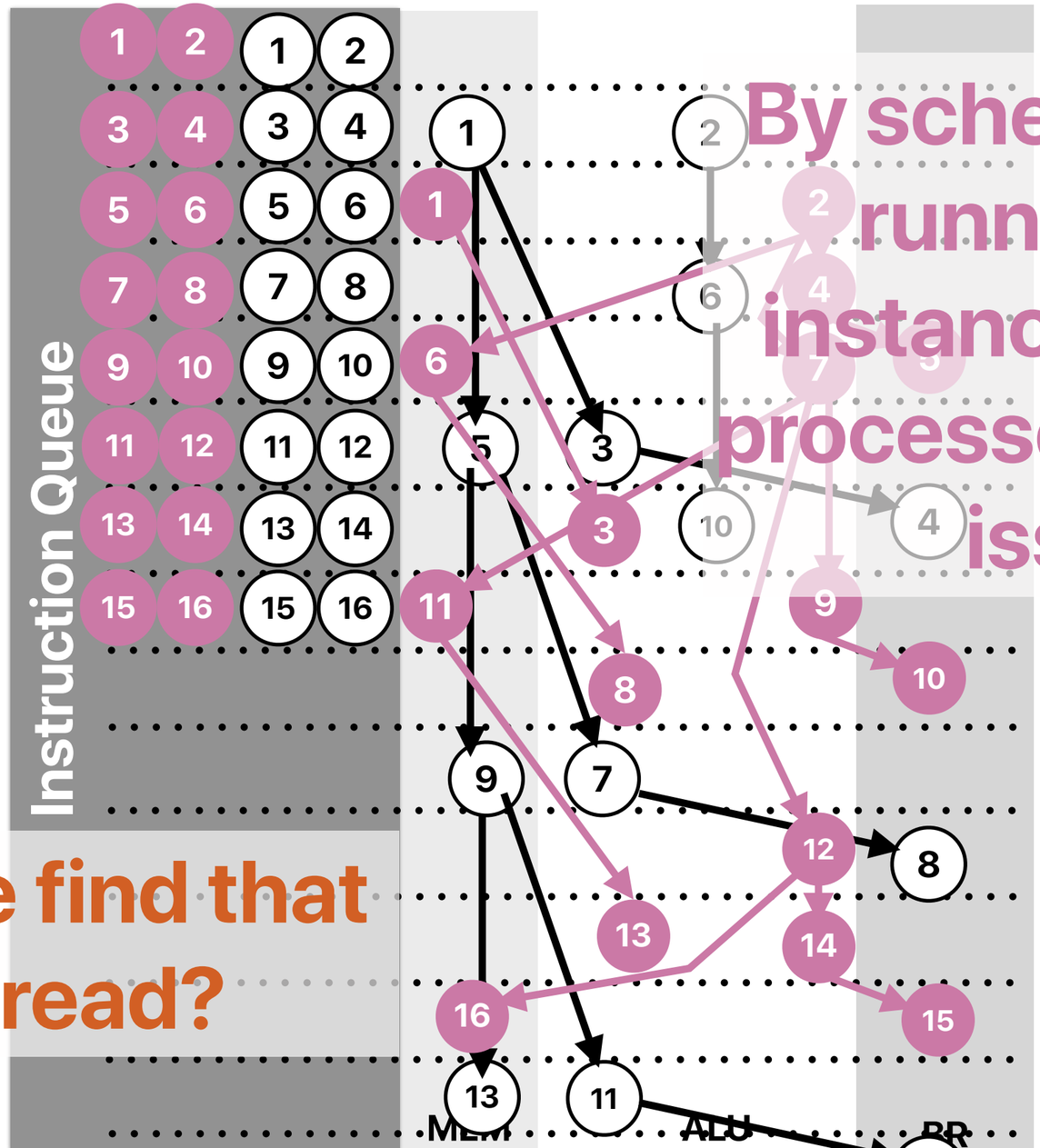
Top at 4.3

Best performing one at 2.7

Performance code may not even fully utilize FUs, either.

Concept: Simultaneous Multithreading (SMT)

```
① movq    8(%rdi), %rdi
② addl    $1, %eax
③ testq   %rdi, %rdi
④ jne     .L3
⑤ movq    8(%rdi), %rdi
⑥ addl    $1, %eax
⑦ testq   %rdi, %rdi
⑧ jne     .L3
⑨ movq    8(%rdi), %rdi
⑩ addl    $1, %eax
⑪ testq   %rdi, %rdi
⑫ jne     .L3
```



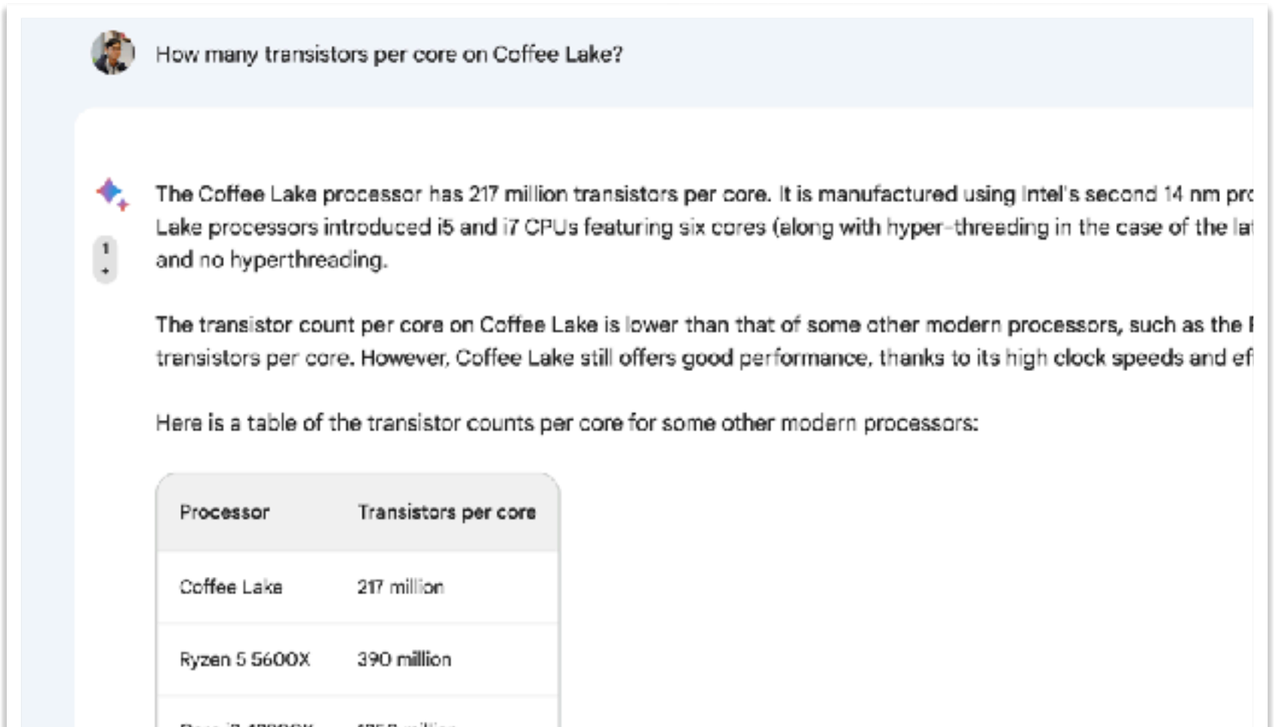

```
① movl (%rdi), %ecx
② cmpq %rdx, %rdi
③ addl %ecx, %eax
④ jne  .L3
⑤ movl (%rdi), %ecx
⑥ addq $4, %rdi
⑦ addl %ecx, %eax
⑧ cmpq %rdx, %rdi
⑨ jne  .L3
⑩ movl (%rdi), %ecx
⑪ addq $4, %rdi
⑫ addl %ecx, %eax
⑬ cmpq %rdx, %rdi
⑭ jne  .L3
⑮ movl (%rdi), %ecx
⑯ addq $4, %rdi
```

By scheduling another running program instance (thread), the processor has 0 wasted issue slots!

Where can we find that "other" thread?

Recap: One powerful core or two “good enough” cores?

Microarchitecture	Transistor Count	Issue-width	Year
Alder Lake	325 M	5x ALU, 7x Memory	2021
Coffee Lake	217 M	4x ALU, 4x Memory	2017
Sandy Bridge	290 M	3x ALU, 3x Memory	2011
Nehalem	182.75 M	3x ALU, 3x Memory	2008



2x 3-issue ALUs Nehalem

Nehalem 6-issue | Alder Lake 12-issue | Nehalem 6-issue

1x 5-issue ALUs Alder Lake

Based on https://en.wikipedia.org/wiki/Transistor_count

SMT v.s. CMP

- An SMT processor is basically a SuperScalar processor with multiple instruction front-end. Assume within the same chip area, we can build an SMT processor supporting 4 threads, with 6-issue pipeline, 64KB cache or — a CMP with 4x 2-issue pipeline & 16KB cache in each core. Please identify how many of the following statements are/is correct when running programs on these processors.
  - ① If we are just running one program in the system, the program will perform better on an SMT processor — you have more resources for the program
  - ② If we are running 4 applications simultaneously, the cache miss rates will be higher in the SMT processor
  - ③ If we are running 4 applications simultaneously, the branch mis-prediction will be higher in the SMT processor — it depends!
  - ④ If we are running one program with 4 parallel threads, the cache miss rates will be higher in the SMT processor — it depends! — it depends!
  - ⑤ If we are running one program with 4 parallel threads simultaneously, the branch mis-prediction will be longer in the SMT processor — it depends!

  A. 1
  B. 2
  C. 3
  D. 4
  E. 5

There is no clear win on each — why not having both?

```
Architecture:                    x86_64
CPU op-mode(s):                  32-bit, 64-bit
Byte Order:                      Little Endian
Address sizes:                   39 bits physical, 48 bits virtual
CPU(s):                          8
On-line CPU(s) list:             0-7
Thread(s) per core:              2
Core(s) per socket:              4
Socket(s):                       1
NUMA node(s):                    1
Vendor ID:                       GenuineIntel
CPU family:                      6
Model:                           151
Model name:                      12th Gen Intel(R) Core(TM) i3-12100F
Stepping:                        5
CPU MHz:                         3300.000
CPU max MHz:                     5500.0000
CPU min MHz:                     800.0000
BogoMIPS:                        6604.80
Virtualization:                  VT-x
L1d cache:                       192 KiB
L1i cache:                       128 KiB
```

Modern processors have both CMP/SMT

Recap: parallel architectures

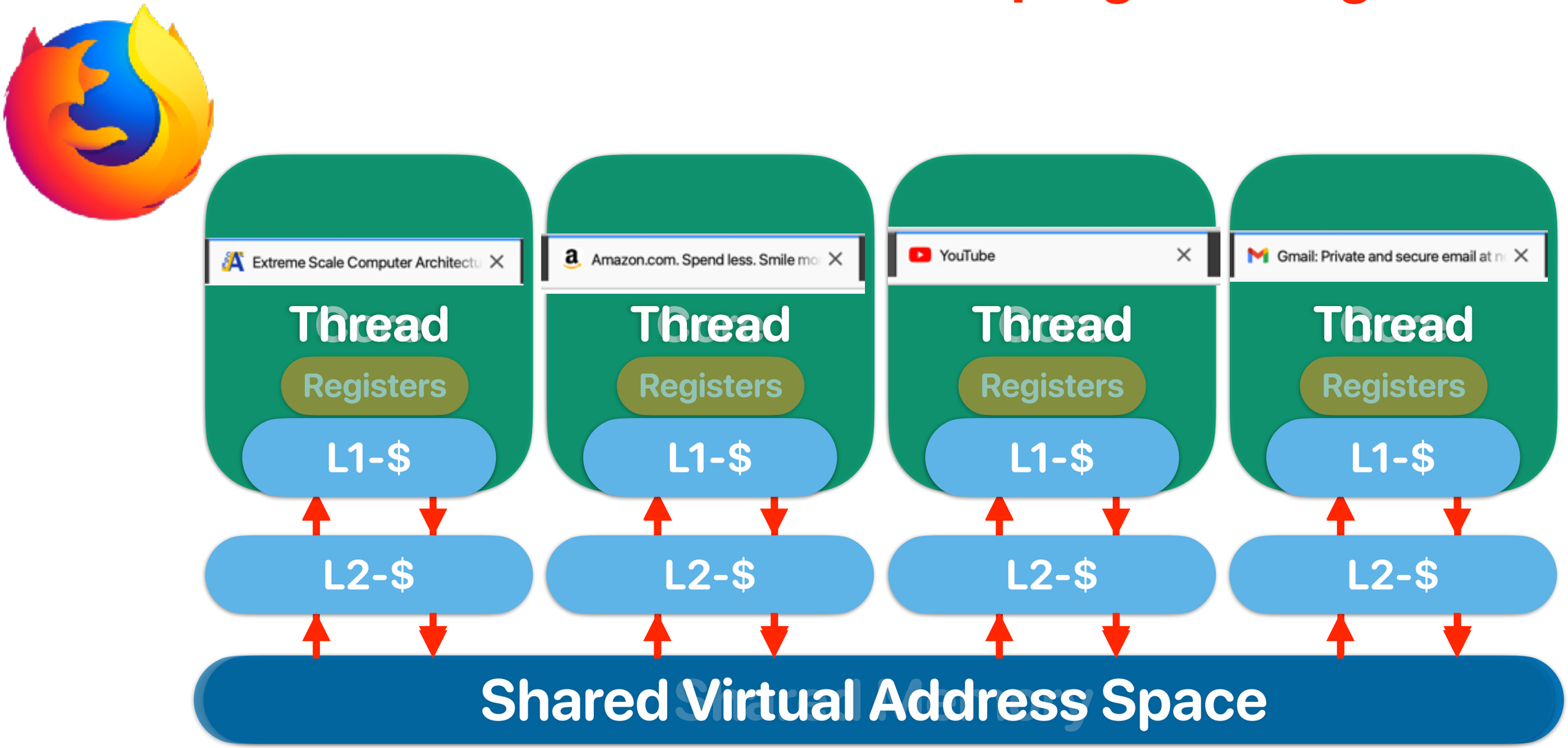
- SMT processors can better utilize the pipeline resources by allowing simultaneous execution of multiple threads
  - Improved execution throughput
  - May hurt the latency of each thread since we share functional units & cache
- CMP provides more processor cores for parallel threads or multiprogrammed environments
  - More isolated, protected pipeline
  - No flexibility if the running program needs more resource

Parallel Programming & Architectural Supports for Parallel Programming

Parallel programming

- To exploit parallelism you need to break your computation into multiple “processes” or multiple “threads”
- Processes (in OS/software systems)
  - Separate programs actually running (not sitting idle) on your computer at the same time.
  - Each process will have its own virtual memory space and you need explicitly exchange data using inter-process communication APIs
  - Programming model: MPI, Spark
- Threads (in OS/software systems)
  - Independent portions of your program that can run in parallel
  - All threads share the same virtual memory space
  - Programming model: pthread or openmp
- We will refer to these collectively as “threads”
  - A typical user system might have 1-8 actively running threads.
  - Servers can have more if needed (the sysadmins will hopefully configure it that way)

What software thinks about “multiprogramming” hardware



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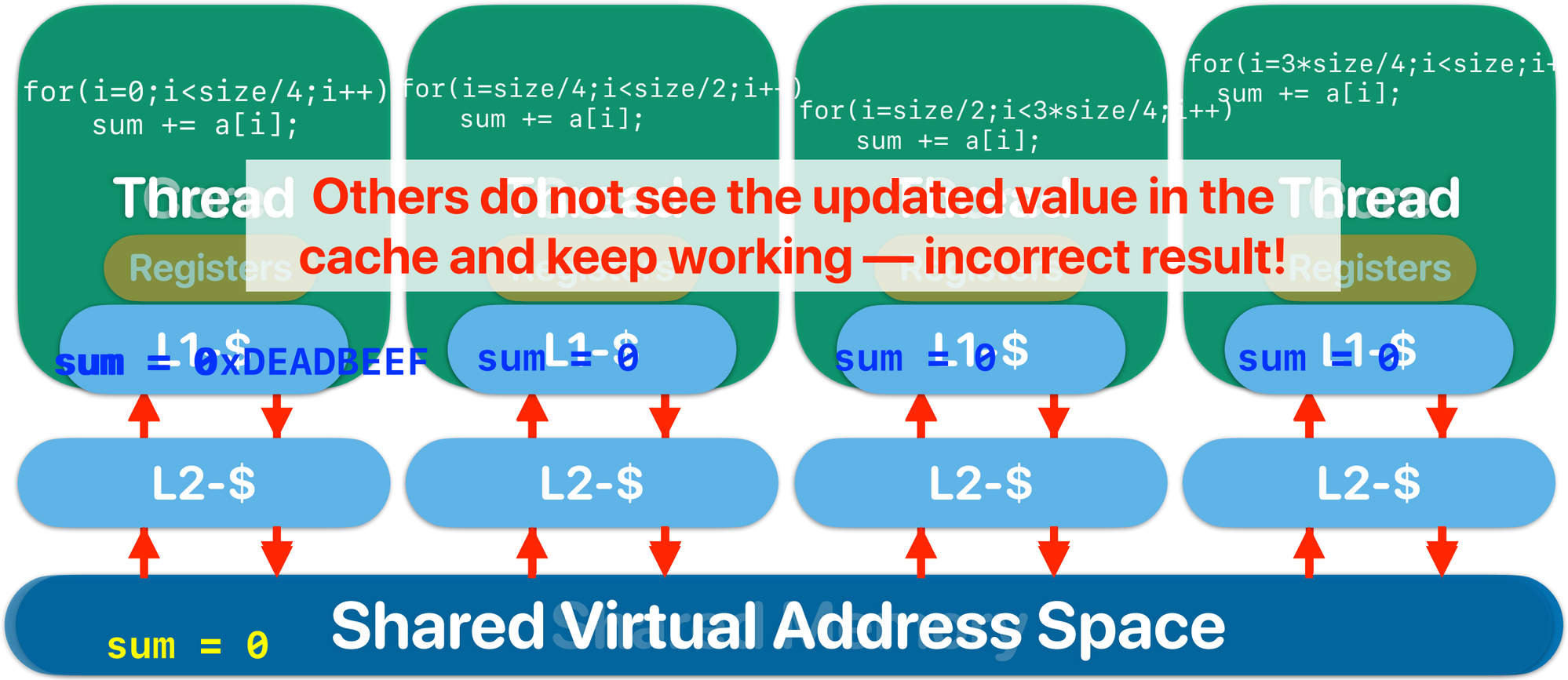


Others do not see the updated value in the cache and keep working — incorrect result!

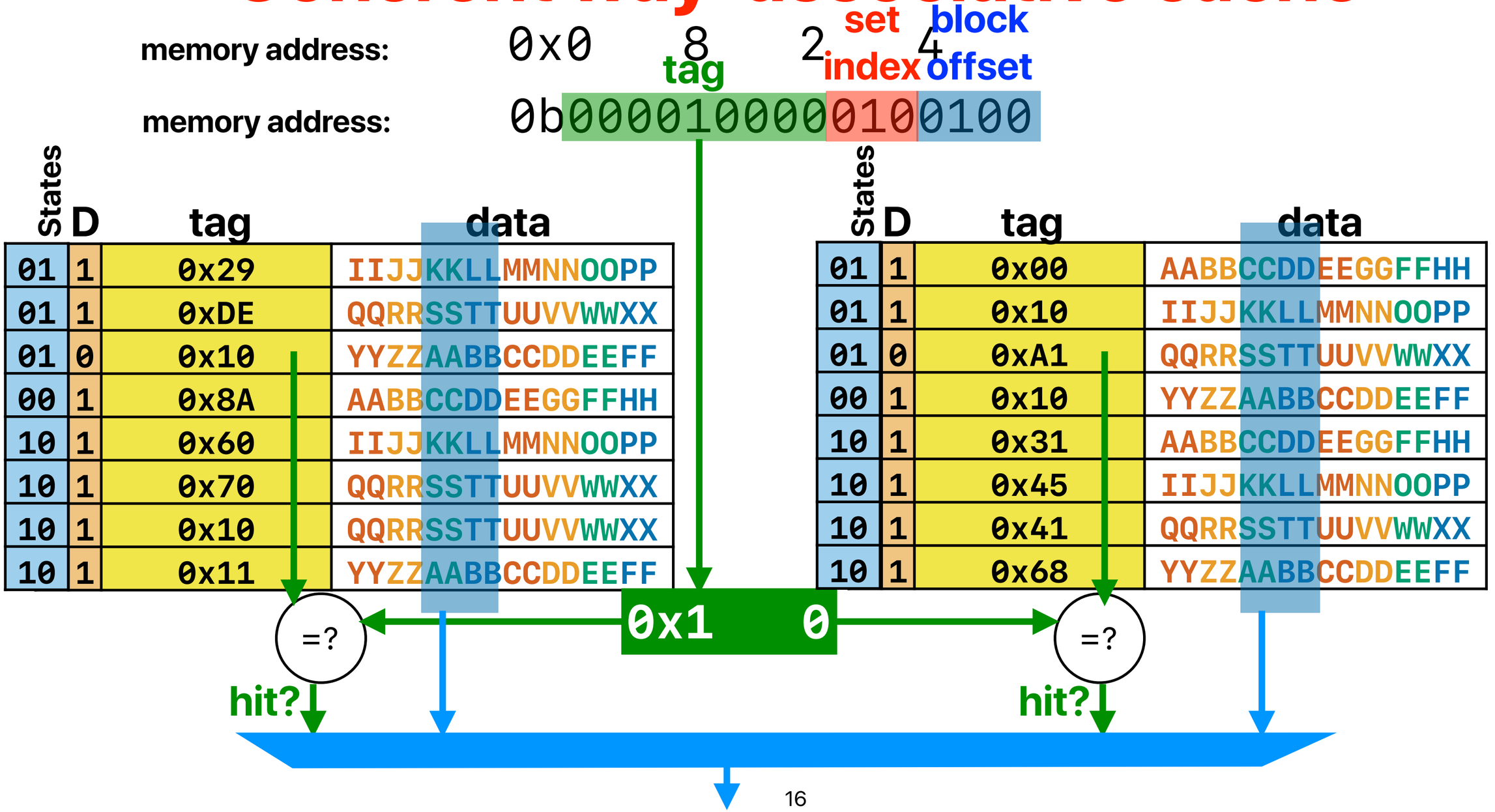
Coherency & Consistency

- Coherency — Guarantees all processors see the same value for a variable/memory address in the system when the processors need the value at the same time
  - What value should be seen
- Consistency — All threads see the change of data in the same order
  - When the memory operation should be done

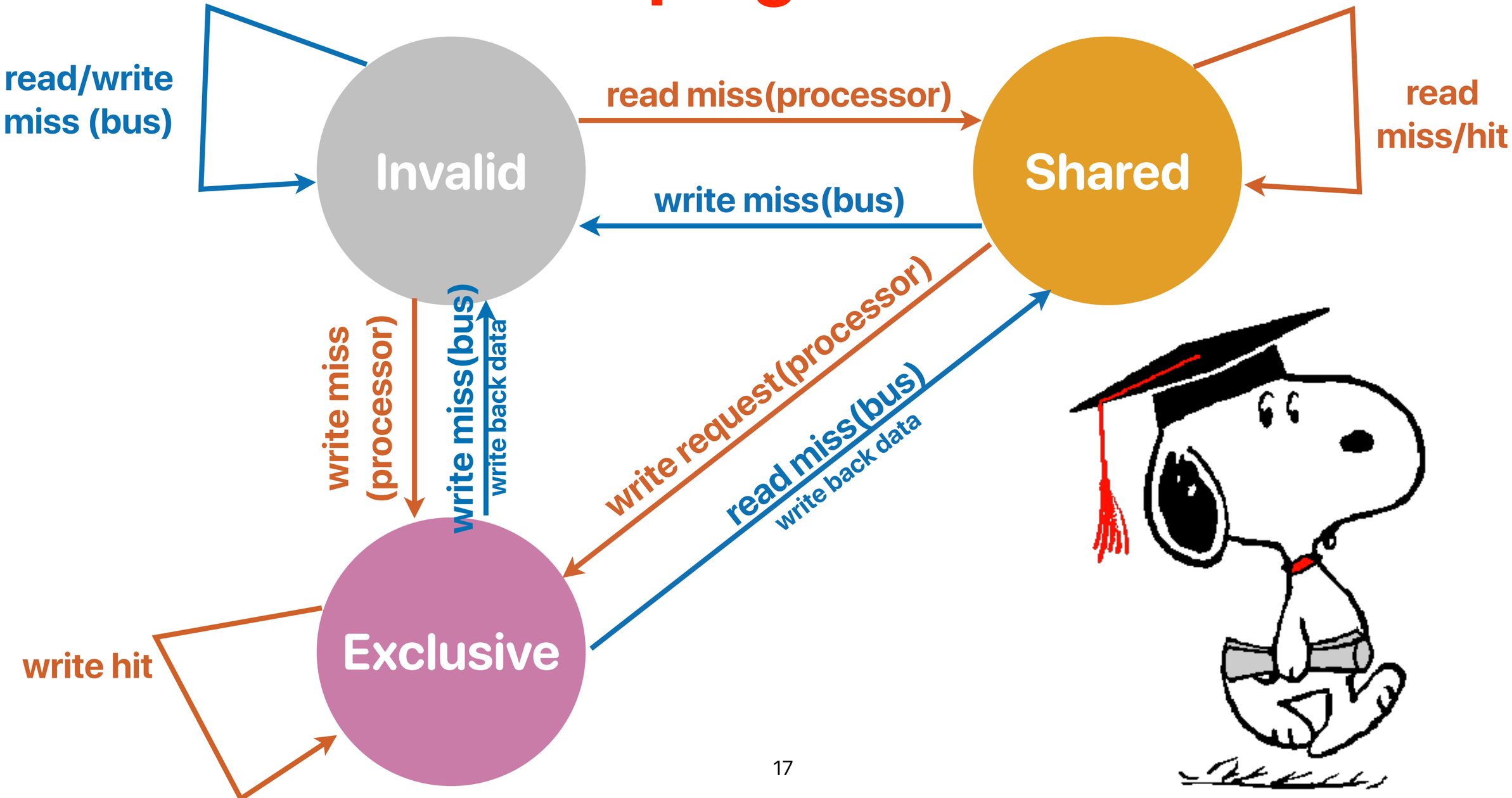
Simple cache coherency protocol

- Snooping protocol
  - Each processor broadcasts / listens to cache misses
- State associate with each block (cacheline)
  - Invalid
    - The data in the current block is invalid
  - Shared
    - The processor can read the data
    - The data may also exist on other processors
  - Exclusive
    - The processor has full permission on the data
    - The processor is the only one that has up-to-date data

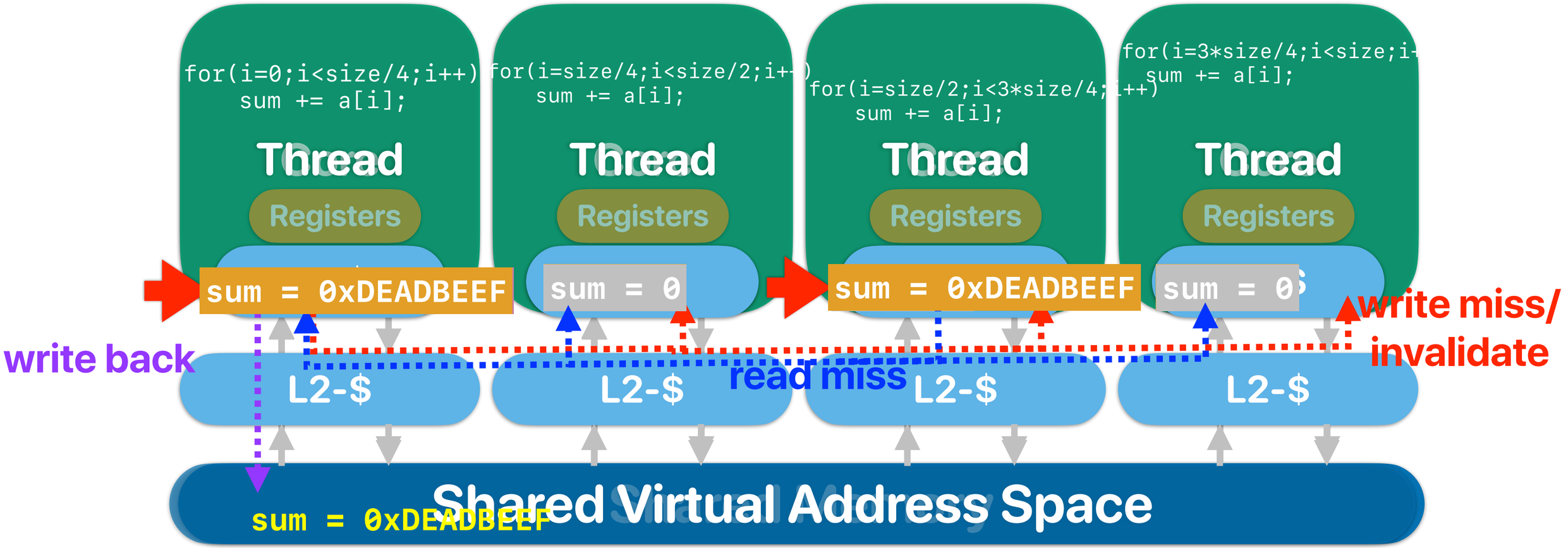
Coherent way-associative cache



Snooping Protocol



What happens when we write in coherent caches?



Observer

thread 1

```
int loop;

int main()
{
  pthread_t thread;
  loop = 1;

  pthread_create(&thread, NULL,
modifyloop, NULL);
  while(loop == 1)
  {
    continue;
  }
  pthread_join(thread, NULL);
  fprintf(stderr,"User input: %d\n",
```

thread 2

```
void* modifyloop(void *x)
{
  sleep(1);
  printf("Please input a number:\n");
  scanf("%d",&loop);
  return NULL;
}
```

Observer

prevents the compiler from putting the variable “loop" in the “register”

thread 1

```
volatile int loop;

int main()
{
  pthread_t thread;
  loop = 1;

  pthread_create(&thread, NULL,
modifyloop, NULL);
  while(loop == 1)
  {
    continue;
  }
  pthread_join(thread, NULL);
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```

thread 2

```
void* modifyloop(void *x)
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  sleep(1);
  printf("Please input a number:\n");
  scanf("%d",&loop);
  return NULL;
}
```

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Cache coherency

- Assuming that we are running the following code on a CMP with a cache coherency protocol, how many of the following outputs are possible? (a is initialized to 0 as assume we will output more than 10 numbers)

thread 1	thread 2
`while(1)` `  printf("%d ",a);`	`while(1)` `  a++;`

① 0 1 2 3 4 5 6 7 8 9
② 1 2 5 9 3 6 8 10 12 13
③ 1 1 1 1 1 1 1 64 100
④ 1 1 1 1 1 1 1 1 1 100

A. 0
B. 1
C. 2
D. 3
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Takeaways: parallel architectures

- SMT processors can better utilize the pipeline resources by allowing simultaneous execution of multiple threads
  - Improved execution throughput
  - May hurt the latency of each thread since we share functional units & cache
- CMP provides more processor cores for parallel threads or multiprogrammed environments
  - More isolated, protected pipeline
  - No flexibility if the running program needs more resource
- Cache coherence can guarantee that everyone would eventually have a coherent view of data, but not when

Take-aways: parallel programming

- Cache coherency only guarantees that everyone would eventually have a coherent view of data, but not when

Performance comparison

- Comparing implementations of thread_vadd — L and R, please identify which one will be performing better and why

Version L

```
void *threaded_vadd(void *thread_id)
{
  int tid = *(int *)thread_id;
  int i;
  for(i=tid;i<ARRAY_SIZE;i+=NUM_OF_THREADS)
  {
        c[i] = a[i] + b[i];
  }
  return NULL;
}
```

Version R

```
void *threaded_vadd(void *thread_id)
{
  int tid = *(int *)thread_id;
  int i;
  for(i=tid*(ARRAY_SIZE/NUM_OF_THREADS);i<(tid+1)*(ARRAY_SIZE/NUM_OF_THREADS);i++)
  {
      c[i] = a[i] + b[i];
  }
  return NULL;
}
```

A. L is better, because the cache miss rate is lower
B. R is better, because the cache miss rate is lower
C. L is better, because the instruction count is lower
D. R is better, because the instruction count is lower
E. Both are about the same

Main thread

```
for(i = 0 ; i < NUM_OF_THREADS ; i++)
{
  tids[i] = i;
  pthread_create(&thread[i], NULL, threaded_vadd, &tids
}
for(i = 0 ; i < NUM_OF_THREADS ; i++)
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Version R

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void *threaded_vadd(void *thread_id)
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  int tid = *(int *)thread_id;
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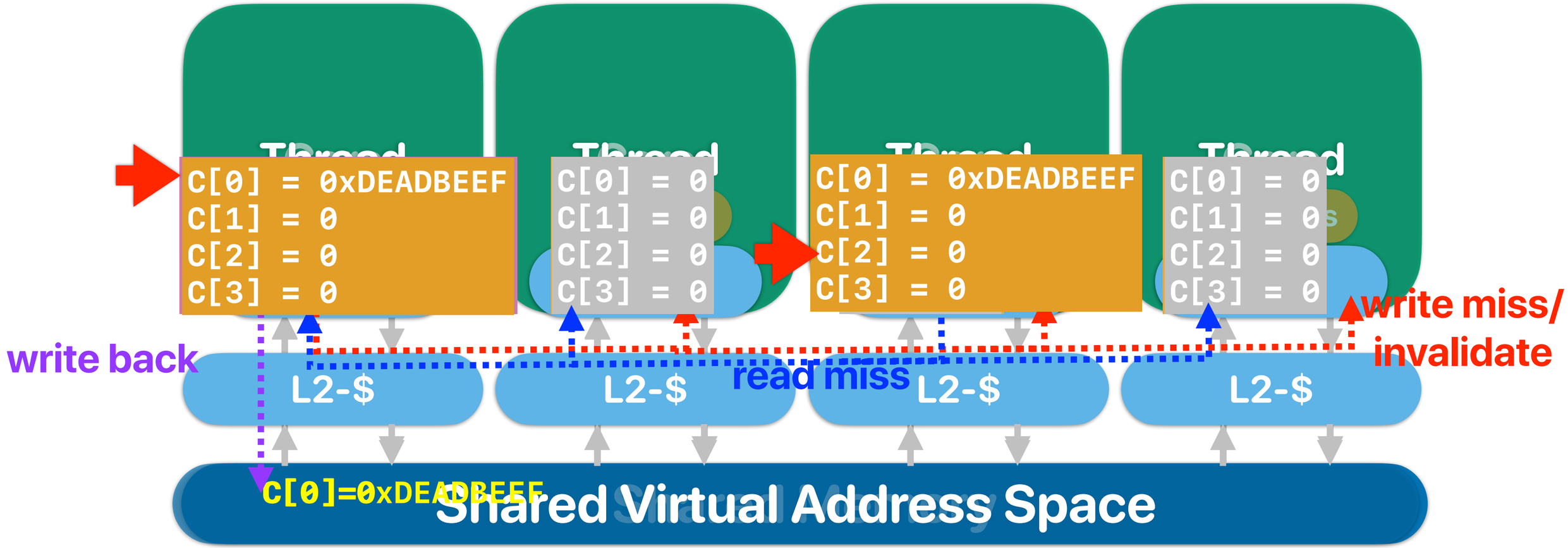
```

Cache coherency



L v.s. R

Version L

```
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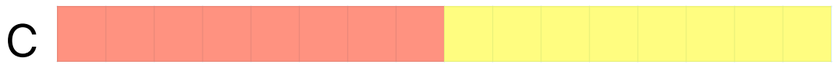
}
```



4Cs of cache misses

- 3Cs:
  - Compulsory, Conflict, Capacity
- Coherency miss:
  - A “block” invalidated because of the sharing among processors.

False sharing

- True sharing
  - Processor A modifies X, processor B also want to access X.
- False sharing
  - Processor A modifies X, processor B also want to access Y. However, Y is invalidated because X and Y are in the same block!

Performance comparison

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Main thread

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  tids[i] = i;
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}
for(i = 0 ; i < NUM_OF_THREADS ; i++)
  pthread_join(thread[i], NULL);
```

Take-aways: parallel programming

- Cache coherency only guarantees that everyone would eventually have a coherent view of data, but not when
- Cache coherency may create unexpected cache invalidations/ misses if you do it wrong

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Again — how many values are possible?

- Consider the given program. You can safely assume the caches are coherent. How many of the following outputs will you see?

① (0, 0)
② (0, 1)
③ (1, 0)
④ (1, 1)

A. 0
B. 1
C. 2
D. 3
E. 4

```c
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <unistd.h>

volatile int a,b;
volatile int x,y;
volatile int f;
void* modifya(void *z) {
  a=1;
  x=b;
  return NULL;
}
void* modifyb(void *z) {
  b=1;
  y=a;
  return NULL;
}
```

```c
int main() {
  int i;
  pthread_t thread[2];
  pthread_create(&thread[0], NULL, modifya, NULL);
  pthread_create(&thread[1], NULL, modifyb, NULL);
  pthread_join(thread[0], NULL);
  pthread_join(thread[1], NULL);
  fprintf(stderr,"(%d, %d)\n",x,y);
  return 0;
}
```

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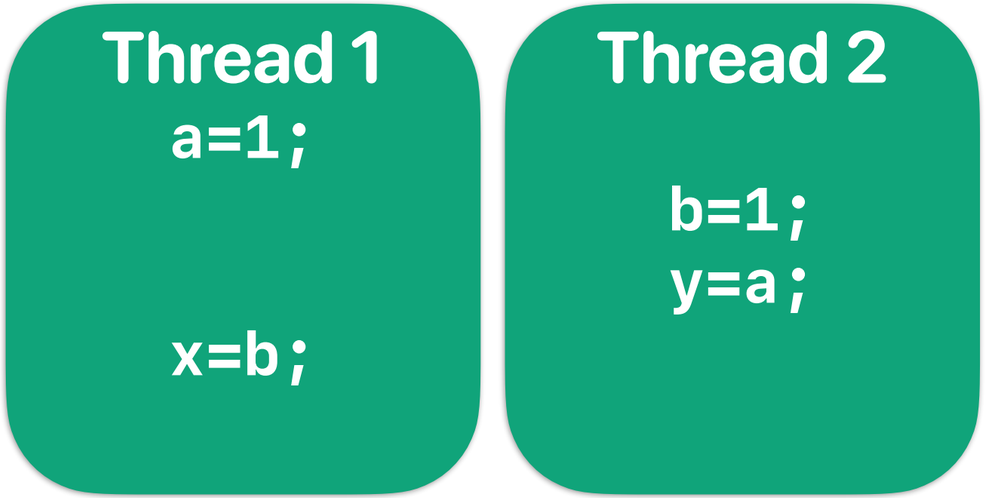
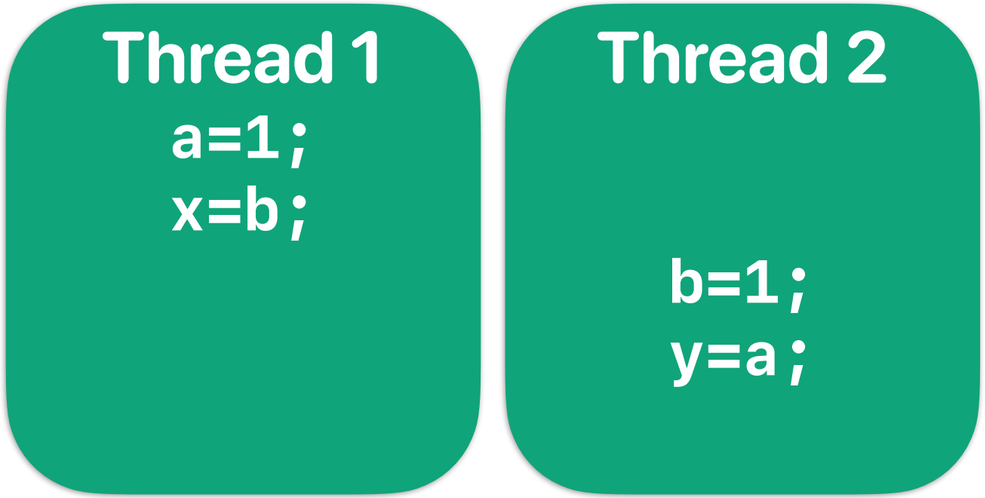
}
```

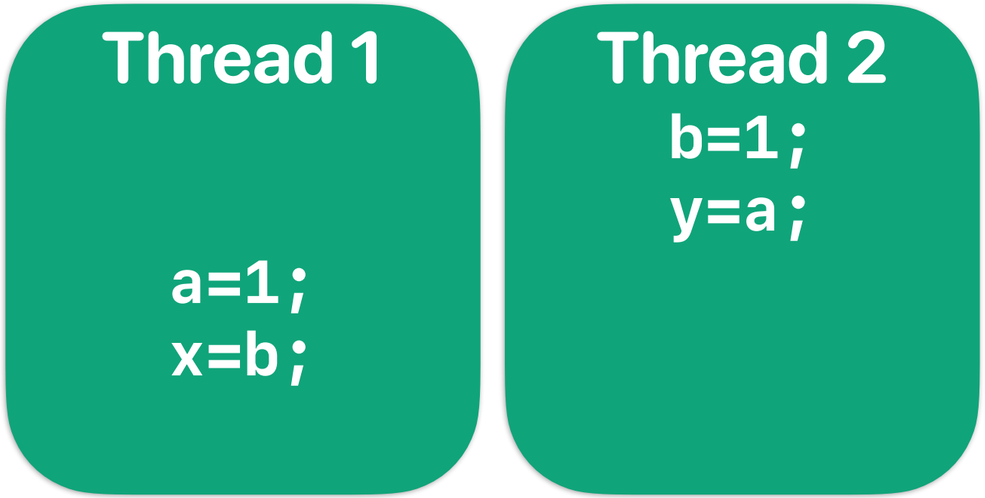
Possible scenarios



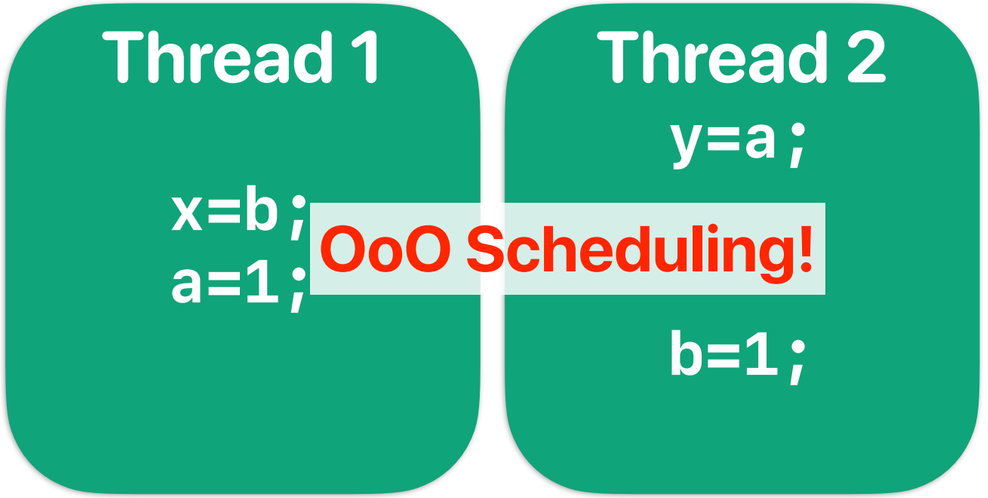
(1,1)



(0,1)



(1,0)



(0,0)

Why (0,0)?

- Processor/compiler may reorder your memory operations/ instructions
  - Coherence protocol can only guarantee the update of the same memory address
  - Processor can serve memory requests without cache miss first
  - Compiler may store values in registers and perform memory operations later
- Each processor core may not run at the same speed (cache misses, branch mis-prediction, I/O, voltage scaling and etc..)
- Threads may not be executed/scheduled right after it's spawned

Again — how many values are possible?

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}
```

Take-aways: parallel programming

- Cache coherency only guarantees that everyone would eventually have a coherent view of data, but not when
- Cache coherency may create unexpected cache invalidations/ misses if you do it wrong
- Processor behaviors are non-deterministic
  - You cannot predict which processor is going faster
  - You cannot predict when OS is going to schedule your thread
  - You cannot predict when the processor is going to schedule an instruction

fence instructions

- x86 provides an “mfence” instruction to prevent reordering across the fence instruction
  - All updates prior to mfence must finish before the instruction can proceed
- x86 only supports this kind of “relaxed consistency” model. You still have to be careful enough to make sure that your code behaves as you expected

thread 1	thread 2

`a=1;`
`mfence` **a=1 must occur/update before mfence**
`x=b;`
	`b=1;`
	`mfence` **b=1 must occur/update before mfence**
	`y=a;`

Take-aways: parallel programming

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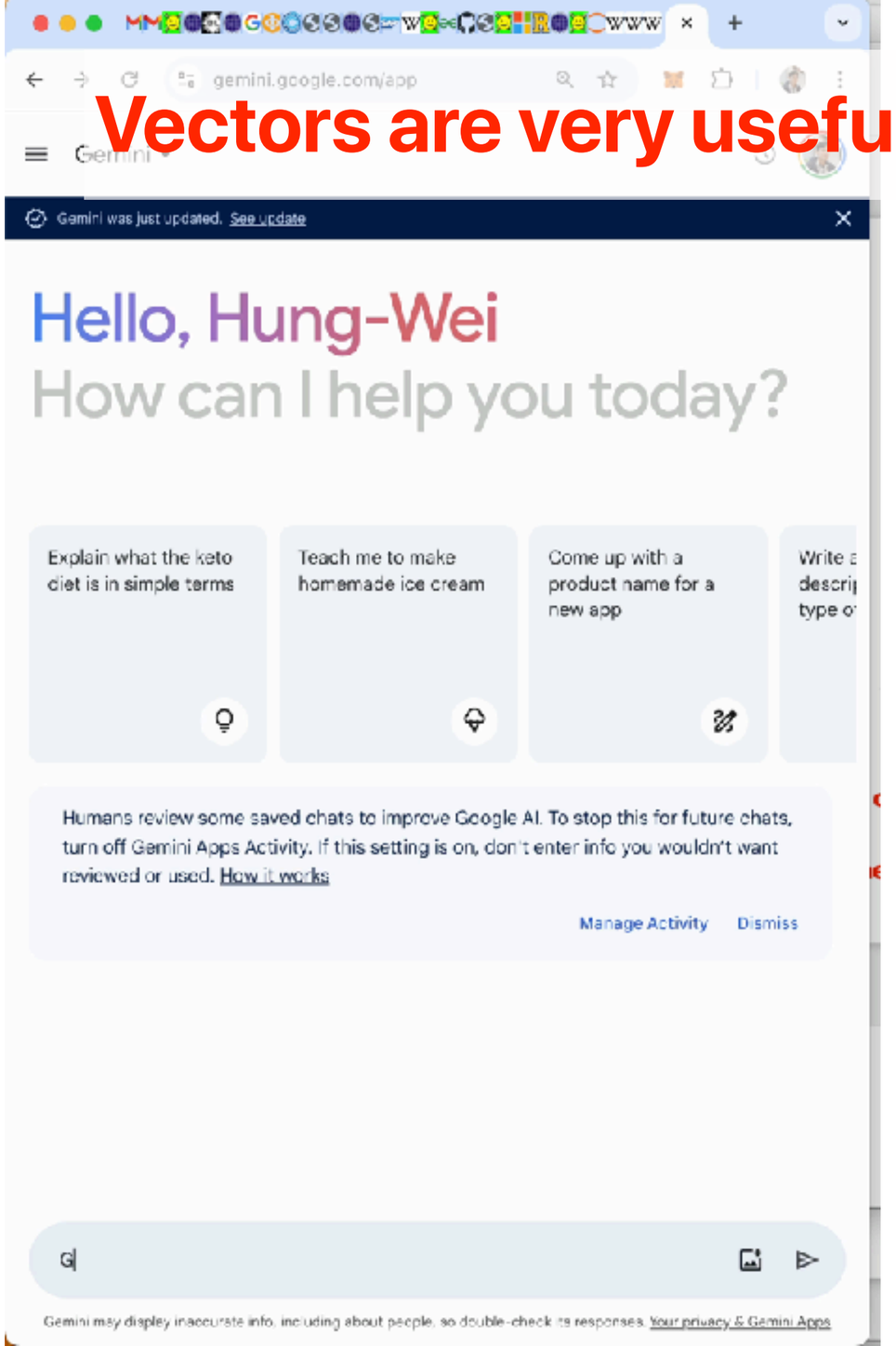
- Processor behaviors are non-deterministic
  - You cannot predict which processor is going faster
  - You cannot predict when OS is going to schedule your thread
  - You cannot predict when the processor is going to schedule an instruction
- Cache consistency is hard to support

Beyond “scalar”

Vectors are very useful data/mathematical representations



- But how do we process them using "superscalar" processors?

```
for(uint64_t i = 0; i < m; i++) {
    result = 0;
    for(uint64_t j = 0; j < n; j++) {
        result += matrix[i][j]*vector[j];
    }
     output[i] = result;
}
```

How well does vector processing map to super "scalar" processors

```
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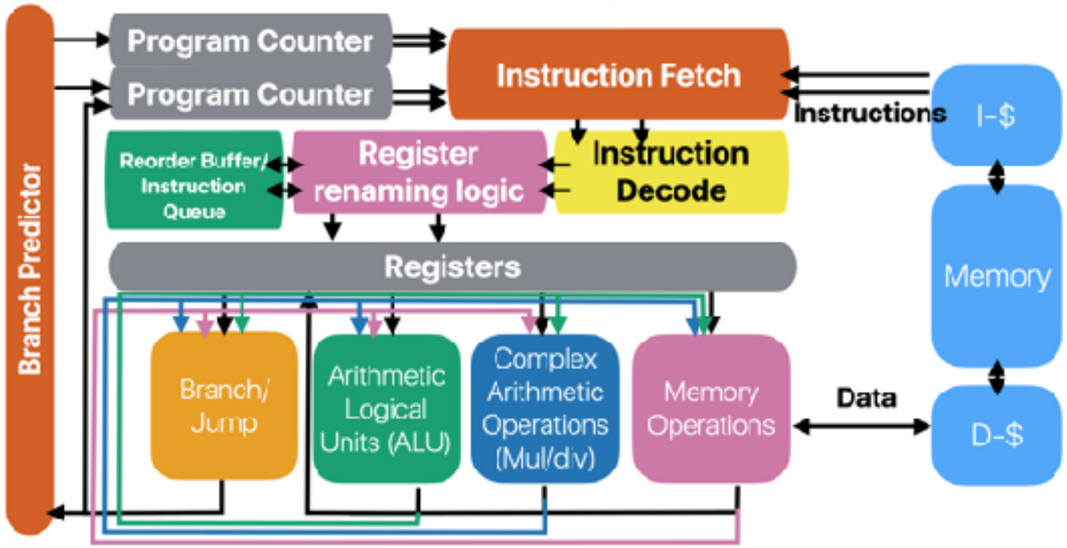

Assume we have a 5-issue INT, 3-load, 4-store pipeline like Alder Lake

Performance is very limited by both the memory bandwidth and available functional units

```
load vector[0]      load matrix[0][1]          load vector[3]          load matrix[0][4]
load matrix[0][0]   load vector[2]             load matrix[0][3]       load vector[5]
load vector[1]      load matrix[0][2]          load vector[4]          load matrix[0][5]

                    mul matrix[0][0], vector[0] mul matrix[0][1], vector[1] mul matrix[0][3], vector[3]
                                                mul matrix[0][2], vector[2]
```

Characteristics of vector processing

- Their operations are uniform across all pairs of elements

— Can we have just one instruction to control all pair-wise operations?

- There are very limited vector operations with mathematical meanings

— Can we simplify the processing elements design and make space for more PEs?

- They have very good spatial locality

— Can we have fetch them once & put in wide registers?

```
for(uint64_t i = 0; i < m; i++) {
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    for(uint64_t j = 0; j < n; j++) {
          result += matrix[i][j]*vector[j];
    }
     output[i] = result;
}
```

```
load vector[0]
load matrix[0][0]
load vector[1]
```

```
load matrix[0][1]
load vector[2]
load matrix[0][2]

mul matrix[0][0], vector[0]
```

```
load vector[3]
load matrix[0][3]
load vector[4]

mul matrix[0][1], vector[1]
mul matrix[0][2], vector[2]
```

```
load matrix[0][4]
load vector[5]
load matrix[0][5]

mul matrix[0][3], vector[3]
```

https://www.pollev.com/hungweitseng close in 1:30

What if we can process a pair of vectors in one instruction?

- If we provide and support vector instructions that can take a pair of vectors as inputs and produce an output in the form of a vector in hardware, how many of the following statement(s) is/are true for the aforementioned matrix-vector multiplication example
  - ① These instructions can help reduce the dynamic instruction count
  - ② These instructions can improve the CPI of the program
  - ③ These instructions can improve the cache miss rate
  - ④ These instructions can improve the instruction-level parallelism
  - A. 0
  - B. 1
  - C. 2
  - D. 3
  - E. 4

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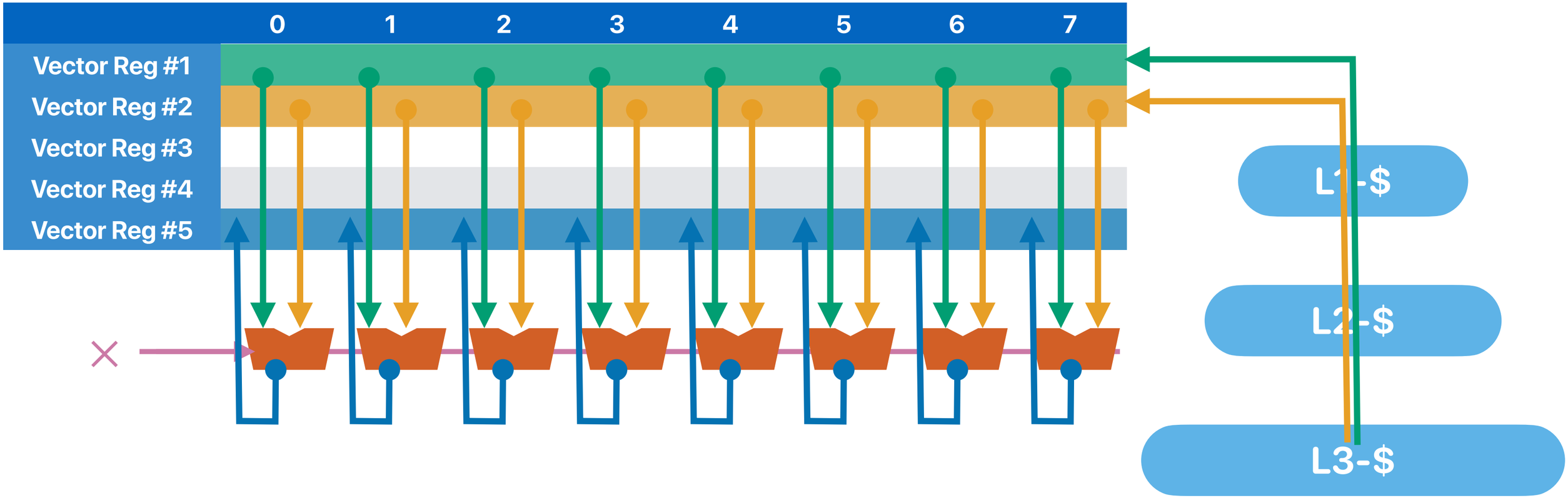
  - A. 0
  - B. 1
  - C. 2
  - D. 3
  - E. 4

— Fewer instructions, but maybe more cycles per instruction

— You still fetch and access by blocks, and in fact, could be worse as you have fewer instructions now

— You don't execute more instructions in parallelism, but only process more data at the same time— data-level parallelism

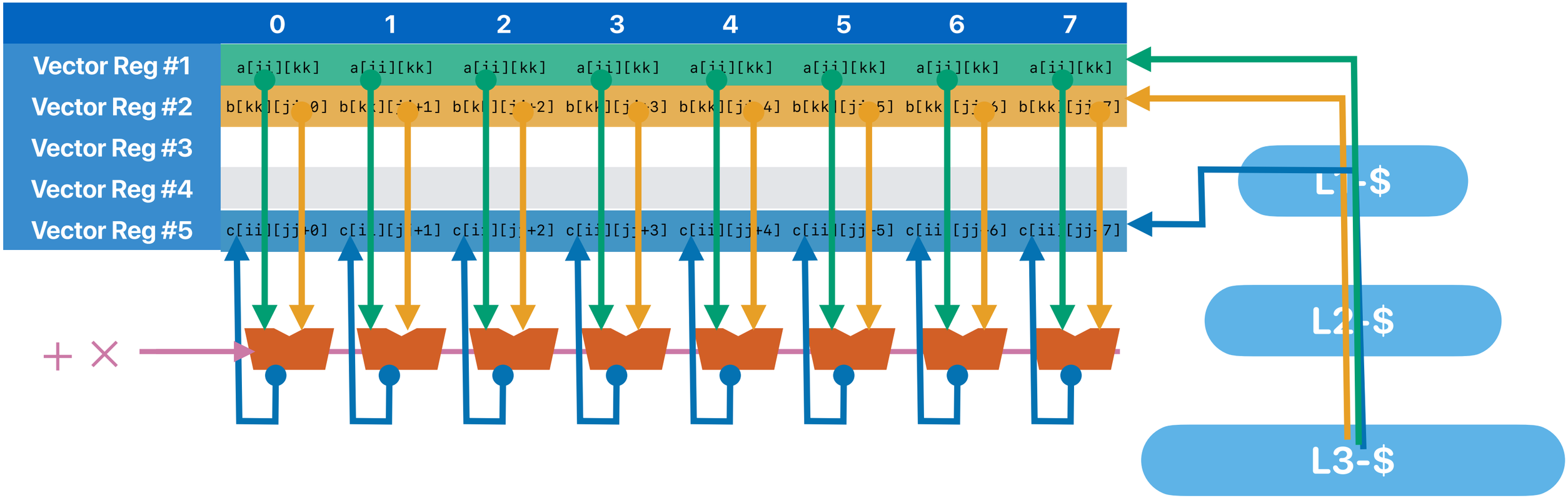
Vector processing architecture



Vectorized matrix multiplications

```
#define VECTOR_WIDTH 8 // VECTOR_WIDTH is 8 as we're using AVX-512
void vector_blockmm(double **a, double **b, double **c, \
uint64_t tile_size) {
  int i,j,k, ii, jj, kk, x;
  __m512d va, vb, vc;
  for(i = 0; i < ARRAY_SIZE; i+=tile_size) {
    for(j = 0; j < ARRAY_SIZE; j+=tile_size) {
      for(k = 0; k < ARRAY_SIZE; k+=tile_size) {
        for(ii = i; ii < i+tile_size; ii++) {
          for(jj = j; jj < j+tile_size; jj+=VECTOR_WIDTH) {
             vc = _mm512_load_pd(&c[ii][jj]); // load c[ii][jj] to c[ii][jj+7] to vc[0-7]
             for(kk = k; kk < k+tile_size; kk++) {
                va = _mm512_broadcastsd_pd(&a[ii][kk]);// load a[ii][kk] to va[0-7]
                vb = _mm512_load_pd(&b[kk][jj]);  // load b[kk][jj] to b[kk][jj+7] to vb[0-7]
                vc = _mm512_add_pd(vc,_mm512_mul_pd(va,vb));    // vc += va * vb
             }
             _mm512_store_pd(&c[ii][jj],vc);      // store vc to c[ii][jj] to c[ii][jj+4]
          }
        }
      }
    }
  }
}
```

Vector processing architecture



Summary: Parallelism in modern computers

- Instruction-level parallelism — perform various, independent instructions simultaneously
  - Pipeline
  - OoO/Superscalar
- Data-level parallelism — perform the same operation on multiple data elements in parallel
  - SIMD instructions
- Thread-level parallelism — perform independent computation streams (composed of many instructions or SIMD instructions)
  - Multicore/SMT processors

Announcements

- **Last reading quiz due next Tuesday**
- Ways to get higher reading quiz average — we drop 3 of your lowest reading quzzes if you submit a proof of iEVAL submission (no ratings should be revealed)
- **Assignment 5 is released** and due 3/13
- Programming assignment due 3/13

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